A Comprehensive Survey of ElectroMagnetic Propagation Models

Veluru Sai Anusha, Nithya G K and Sethuraman N Rao

***Abstract*—Path loss is caused by the propagation characteristics and the obstacles between transmitter and receiver due to which signal strength is reduced in wireless communication. Analysis based on path loss models provides good initial estimate of the signal characteristics. Propagation models are used to determine the path attenuation based on parameters such as carrier frequency, transmitter-receiver distance, basestation antenna height, receiver antenna height and terrain profile (e.g. rural, suburban and urban). Propagation models are used for path loss estimation which is important for planning the deployment of base-stations and area of coverage. In this paper, different Propagation models such as Okumura's Model, Hata Model, Longley rice Model, Free Space Propagation Model, COST 231 Extension to Hata Model, COST 231 Walfisch Ikegami Model, Stanford University Interim (SUI) Model and Hata-Okumura Model are surveyed. A comparative analysis of these models is provided which can serve as a valuable guide for selecting the appropriate model for a specific application scenario.**

Index Terms—Okumura's Model, Hata Model, Line of Sight (LoS), COST 231 Model, Stanford University Interim (SUI) Model, Non Line of Sight (NLoS), Path Loss

I. INTRODUCTION

Wireless Communication systems provide good quality, high speed data exchange using portable devices. In wireless technologies, information is delivered by electromagnetic waves. As a result of obstructions like trees, buildings, hills, mountains, the radio waves cannot directly reach the receiver since they are obstructed along the Line of Sight (LoS) path. The loss of signal strength which occurs between basestation and receiver is known as path attenuation. Path attenuation is defined as the difference between transmit power and receive power as expressed below [11]:

P_L (dB) $=P_{TR} + G_{TR} + G_{RE} – P_{RE} - L_{TR} - L_{RE}$

Where P_{TR} and P_{RE} represent transmit power and receive power

G_{TR} and G_{RE} represent gain of transmit and receive antennas

L_{TR} and L_{RE} represent Feeder losses

Fig. 1. Line of Sight (LoS) Communication

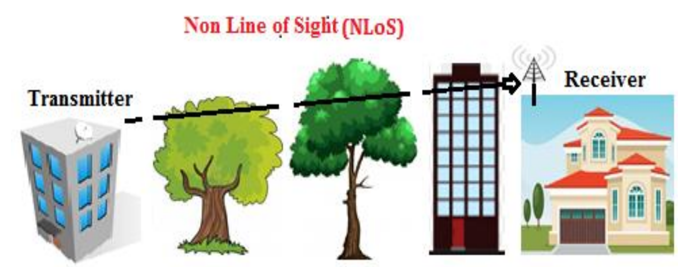

Fig. 2. Non Line of Sight (NLoS) Communication

Path attenuation may be affected by the following mechanisms

- Absorption
- Refraction
- Reflection
- Diffraction
- Scattering

Absorption losses occur if the wave passes through obstacles such as trees, buildings. Refraction loss occurs when an electromagnetic wave travels from one medium to another medium with different refractive indices, results in change of velocity and change of direction. Reflection losses occur when the transmission path between transmitter and receiver is blocked by surfaces such as ground, walls, etc., and from furniture (tables, chairs, desks).

Veluru Sai Anusha is with Amrita Center for Wireless Networks & Applications (AmritaWNA), Amrita School of Engineering, Amritapuri, Amrita Vishwa Vidyapeetham, Amrita University, India (e-mail: anushachowdary966@gmail.com).
Nithya G K is with Amrita Center for Wireless Networks & Applications (AmritaWNA), Amrita School of Engineering, Amritapuri, Amrita Vishwa Vidyapeetham, Amrita University, India (e-mail: nithyagk@am.amrita.edu).
Sethuraman N Rao is with Amrita Center for Wireless Networks & Applications (AmritaWNA), Amrita School of Engineering, Amritapuri, Amrita Vishwa Vidyapeetham, Amrita University, India (e-mail: sethuramanrao@am.amrita.edu).

978-1-5090-3800-8/17/$31.00 ©2017 IEEE

Diffraction loss occurs when there is an obstacle in the transmission path and signal bends around an object. Scattering is a physical procedure where moving particles, sound or light are made to diverge from a direct path by one or more limited non-uniformities in the medium through which they travel. Scattered waves are caused by obstacles and irregular surfaces [1].

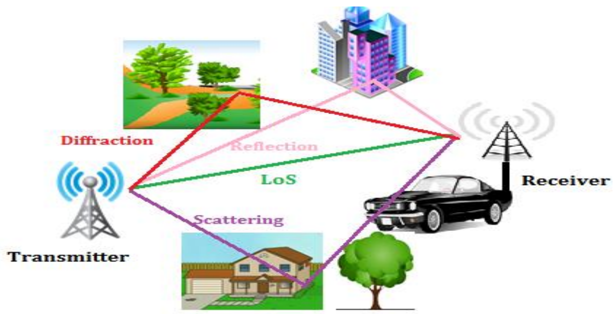

Fig. 3. Phenomena of Diffraction, Reflection and Scattering

For wireless communication systems, propagation path loss models play a key role in determining the signal strength, outdoors as well as indoors. Path loss can be calculated in different terrain profiles such as rural, sub-urban and urban with the help of path loss models. During the beginning stage of the deployment, path loss models are used for conducting practical studies. As the deployment proceeds, propagation models are also helpful for preforming interference studies. Some of the technical challenges of path loss modelling are accounting for spectrum limitations, user mobility and multipath propagation. Selection of an appropriate propagation path loss model is the first step in the network design. Propagation path loss models can be classified into two categories:

i) Indoor Propagation models
ii) Outdoor Propagation models

Indoor Propagation models are used to estimate the signal strength when the Transmitter-Receiver Separation is small, of the order of a few hundred meters, typically indoors. This model is influenced by particular features such as building type, construction materials and layout of the building. Outdoor Propagation models are used to estimate the signal strength at a particular area and based on standard interpretation of measured data acquired from the environment [4]

Propagation path models represent a set of algorithms and mathematical equations that are used for signal strength estimation in a particular terrain profile. Propagation models can be classified into three types of models i.e. Empirical models, Deterministic models and Statistical models.

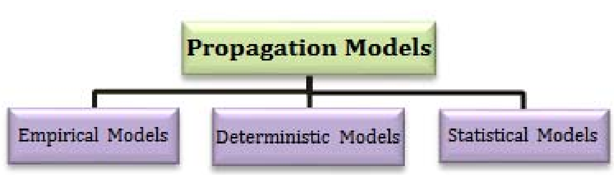

Fig. 4. Classification of Propagation Models

Empirical models use a set of equations obtained from results of several measurements. These models give accurate results and are used more often in practical applications when compared to Deterministic and Statistical models. Deterministic Models use reflection & diffraction laws which govern electromagnetic signal propagation to calculate the received signal strength at a particular location and need an entire 3-Dimensional map of the propagation area. The deterministic models may require computational efficiency and algorithms used by these models are extremely complex [1]. Deterministic models are based on fixed geometry (streets, buildings). Statistical models model the terrain profile as a series of random variables and depend on probability analysis to predict path loss. These models need least information about the terrain profile and are least accurate.

There are three different area types, namely, Rural, Sub-Urban and Urban.

Rural Area– no obstacles like big trees or buildings in the signal propagation path, influence of terrain height

Sub-Urban Area – village covered with houses and trees. Interference near the mobile not very overcrowded

Urban Area- large cities with huge buildings of several floors or larger village scattered with thickly-grown tall trees.

Our University's Parent NGO has adopted 101 villages across the various states of India. Research on affordable technology options for rural connectivity is currently underway at our research center [17]. Our center is also conducting research on antennas such as coplanar Pentagonal Antenna, CPW fed Monopole, Dual Beam Pentagonal Patch Antenna, Rectangular Microstrip Antenna and Dielectric Resonator Loaded Microstrip Antenna [6][7][8]. A project to provide internet connectivity to marine fishermen far into the sea is currently in progress [9][10]. This work provides a comparative analysis of various propagation models which can serve as a valuable guide for selecting the appropriate model for a specific application scenario.

The rest of this paper is organized in the following sections. Section II discusses different propagation models for path loss estimation. Section III presents comparative study of path loss models based on parameters such as frequency range, type of terrain, etc. and lists the advantages and disadvantages. The work is concluded in section IV.

II. Description of Propagation Models

This section discusses different empirical path loss models briefly. Some of the models are limited to the lower frequencies. Different propagation path attenuation models are suitable for different types of terrain profile like rural, sub-urban and urban.

A. Okumura Propagation Model

Okumura's model is the simplest and gives best accuracy in path loss estimation in land mobile radio and cellular systems. This model is used for determining the path loss in the frequency range of 150MHz to 1920MHz. Antenna height of base station can be in the range 30-1000m and antenna height of mobile station antenna can be 1-10m [5].

Path loss equation for Okumura model is:

$L_{50\%}$ (dB) = L_{fs} (d) +A_{ma} (f_{re}, d) – G_b (h_{rec}) – G_m (h_{bs}) - G_{AR}

Where $L_{50\%}$ = 50^{th} percentile median value of path loss

L_{fs} (d) = free space propagation loss = 20log (4*PI*d*f/c)

f_{re} = frequency, c= speed of light, d = distance

G_b (h_{te}) = Antenna height of transmitter gain factor

G_m (h_{rec}) = Antenna height of mobile station gain factor

G_{AR} = Correction gain factor (type of environment)

A_{ma} (f, d) = median attenuation in addition to free space model [13]

=13dB from curve @2.4GHz

G_m (h_{rec}) = 20 log (h_r/3)   10m > h_{rec} >3m

= 10 log (h_r/3) h_{rec} < 3m

G_b (h_{bs}) = 20 log (h_{bs}/200)   1000> h_{bs} >30m

= 10 log (h_{bs} /200) h_{bs}< 30m

B. Hata Propagation Model

This model provides many formulas for path loss with various types of environments and carrier frequency ranging from 150MHz to 1.8GHz. The Antenna height of receiver is in the range of 1-10m and the Antenna height of transmitter is in the range of 30-200m. In this model, Link distance is 1-10km. This model is fit for huge-cell mobile systems, but this model is not suited for personal communication.

The path attenuation equation in urban environment is [13]

P_L (urban) = 26.6 log (f_c) – 13.82 log (h_{bs}) –a (h_{rec}) + 69.55 + (44.9-6.55 log (h_{bs})) log (d)

Where a (h_{rec}) = (1.11 log (f)-0.7) h_{rec} – (1.56 log (f_c) – 0.8)

The path attenuation equation in sub-urban environment is

P_L (sub-urban) = P_L (urban) – {log (f_c/28}-5.4

The path loss equation in rural terrain is

P_{Loss} = P_L (urban) – 4.78(log f_c) ^ 2 – 40.94 +18.33 log (f_c)

C. Stanford University Interim (SUI) Propagation Model

Stanford University Interim (SUI) Propagation model is evolved as a part of IEEE 802.16 by Stanford University and is derived from Multipoint Microwave Distribution system. This model uses frequency spectrum between 2.5GHz and 2.7GHz. In this model, Antenna height of base station is between 15m to 40m and Antenna height of receiver is between 2m to 10m. SUI model covers three categories of terrain profile namely Category A, Category B and Category C

Category A – Presents high path loss and suitable for hilly environment with moderate to heavy tree densities [12]

Category B – distinguished with either flat terrains or hilly areas with rare vegetation or flat terrain with high vegetation

Category C – Presents low path loss and useful for flat terrain with moderate to heavy densities [2]

The path attenuation equation is as follows:

P_L = F_s +10γ log [d/d0] + X_{fr} + X_{ha} + S_h for d0<d

Where d = distance between Access Point and Customer Premise Equipment

d0 = reference distance 100(m)

F_s = free-space path loss

γ = Path attenuation exponent value

X_{fr} = frequency correction factor

X_{ha} = Receiving antenna correction factor

Sh = correction factor for shadowing (dB)

The parameter F_s is calculated by

F_s = 20 $\log_{10}$ (4πd0/λ)

γ = a_0-b_0 h_{bs} + (c_0/h_{bs})

γ>5   for indoor propagation

3< γ <5   for urban Non Line of Sight (NLoS) environment

γ=2   for free space path loss

For Terrain Category A and Category B

X_{ha} = -10.8 $\log_{10}$ (h_{re}/2000)

X_{fr} = 6 $\log_{10}$ (f/2000)

For Terrain Category C

X_{ha} = -20 $\log_{10}$ (h_{re}/2000)

Where f is the frequency

h_{re} is the Antenna height of CPE

h_{bs} is the Antenna height of base station

TABLE I

PARAMETERS VALUES FOR SUI IN DIFFERENT TYPES OF TERRAINS

Parameter	Category A	Category B	Category C
a_0	4.6000	4.000	3.600
b_0	0.0075	0.0065	0.05
c_0	12.600	17.10	20.00
S_h	10.6	9.6	8.2

D. COST 231 Extension to Hata Propagation Model

This model is used to determine the path loss for the distance from basestation to receiver upto 20km. This model requires that the transmitter antenna is higher than all rooftops. The Antenna height of transmitter is between 30m to 200m and Antenna height of receiver ranging from 1m to 10m.

The path loss equation in urban environment is

P_{Loss} (dB) = 33.9 log f_c – 13.82 h_{tr} – ah_{re} + (44.9-6.55) log h_{tr}) log d + C_m + 46.3

Correction factor for urban environment is

ah_{re} = 3.20($\log_{10}$ (11.75h_{re})) ^{2}-4.79

Correction factor for sub-urban and rural areas is

ah_{re} = (1.11$\log_{10}$ (f_r)-0.7) h_{re} - (1.5$\log_{10}$ (f_r)-0.8)

C_m = 0dB for sub-urban areas and medium cities

= 3dB for Cosmopolitan areas

E. COST 231 Walfisch-Ikegami Propagation Model

COST 231 W-I Model is the most suitable model for sub-urban and rural terrain profiles which have uniform building height and gives accurate path attenuation estimation. Antenna height of transmitter is between 4 to 50m and Antenna height of receiver is between 1 to 3m

The path attenuation equation for LoS Communication [16] is

PL_{LoS} = 20log (f_c) + 42.6 +26 log (d)

The path loss equation for Non Line of Sight (NLoS) Communication is

PL_{NLoS} = L_{FS} + L_{rts} (f_c, w_r, Δh_{mb}, Φ) + L_{msd} $h_{bs,}$ (Δh_{bs}, d, f_c, b_s)

L_{FS} = free space propagation path loss

L_{RTS} represents diffraction from rooftop to street

$= -10 \log (w) - 16.9 + 10 \log (f_c) + 20 \log (h_{tr}-h_{re}) + L_{orie}$

L_{orie} represents diffraction loss due to multiple obstacles

L_{msd} = multi-diffraction loss

$= L_{bsh} + H_a + H_d \log_{10}[d] + H_f \log_{10}(f\,[MHz]) - 9 \log_{10}(b)$

F. Longley Rice Path Loss Model

Longley Rice Model is useful for Point to Point communication for different kinds of terrain and frequency range is 40MHz to 100MHz. There are two modes of operation in Longley Rice Model.

Point to Point Prediction: If the terrain profile is accessible, particular path parameters are calculated based on some measurements

Area Mode Prediction: If the terrain profile is not accessible, Longley Rice Path loss model provides techniques (geometric optics, Knife-edge diffraction) to calculate certain parameters [4].

G. Free Space Propagation Model

Path attenuation in Free Space model defines the extent of signal strength lost in the time of signal propagation from transmitter to receiver [15]. The path attenuation equation for this model is

$PL\,(dB) = 32.45 + 20 \log_{10}(d) + 20\log_{10}(f_c)$

Where f_c is the frequency (MHz) and d is the Distance between basestation and Customer Premise Equipment (CPE)

H. Two Ray Ground Path Loss Model

Two ray Ground Model is based on geometric optics and considers both ground reflected propagation path and direct path between transmitter and receiver. For mobile radio systems, this model is accurate for predicting the signal strength over larger distances of several kilometers. The formula for two ray ground model is expressed below

$$P_r = \frac{P_t * G_t * G_r * (H_t{}^\wedge 2 * H_r{}^\wedge 2)}{d^n * L}$$

Where L=1 (system loss)

P_t = Transmit Power (w)

P_r = Receive Power (w)

G_t = transmit gain

G_r = receive gain

I. Hata-Okumura Model

Hata-Okumura is a very popular model and can provide predictions upto 2GHz. This model is used by computer simulation tools and quite accurate. This model is useful for large cell coverage.

The path attenuation formula in urban environment is

$P_L\,(dB) = 26.16 \log f_r - 13.82 \log_{10} h_{bs} + (44.9 - 6.55\log_{10} h_{bs})\, R - H + 69.55$

For large cities $f_r \geq 300$MHz, $H = 3.2(\log(11.7554h_{ms}))^2 - 4.97$

For large cities $f_r < 300$MHz, $H = 8.29(\log(1.54h_{ms}))^2 - 1.1$

For medium to small cities $H = (1.1 \log_{10}f_r - 0.7)\, h_{ms} - (1.56 \log f_r - 0.8)$

The path attenuation in sub-urban environment is

$P_L\,(dB) = 26.16 \log f_c - 13.82 \log_{10}h_{bs} + (44.9 - 6.55\log_{10}h_{bs}) \log_{10} R - 2(\log_{10}(f_c/28))^2 + 5.4 + 69.55$

The path loss in rural terrain is

$P_{Loss}\,(dB) = 26.16 \log_{10}f_c - 13.82 \log_{10}h_{bs} + (44.9 - 6.55\log_{10}h_{bs})\, R - 4.78(\log_{10}f_c)^2 + 18.33\log_{10}f_c + 40.94 + 69.55$

Where h_{bs} is the Antenna height of transmitter above height of local terrain

h_{ms} is the Antenna height of receiver above height of local terrain

$R = r \times 10^{-3}$ represents circle distance between basestation and receiver

J. Ericsson 9999 Path Loss Model

Ericsson 9999 path loss model is developed by Ericsson as an enhancement of Hata Model. Ericsson model uses frequency range up to 1900MHz. In this model, adjustment of the factors is possible as per the given environment [1]. The path loss in Ericsson model is calculated as

$PL_{Ericsson} = b_0 + b_1 \log(d) + b_2 \log(h_{te}) + b_3 \log(h_{te}).\log d - 3.2(\log(11.75))^2 + h(f_0)$

Where $h(f_0) = 44.49 \log(f_r) - 4.78 (\log(f_r))^2$

TABLE II

PARAMETERS VALUES OF ERICSSON IN DIFFERENT TYPES OF TERRAINS

Parameters	Rural Area	Sub-Urban Area	Urban Area
b_0	45.95	43.0	36.2
b_1	100.6	68.93	30.2
b_2	12.0	12.0	12.0
b_3	0.1	0.1	0.1

III. COMPARATIVE ANALYSIS OF PROPAGATION MODELS

This section provides a comparative study of different propagation path attenuation models based on factors such as frequency range, type of terrain, advantages and disadvantages.

Based on the comparative analysis, it is found that the SUI model is suitable for all the three regions i.e. rural, sub-urban and urban. Okumura Model is generally used in urban environment. Hata model is a very good model for first generation cellular systems. COST 231 Hata path loss model is suitable for estimating path attenuation in wireless mobile systems. The COST 231 Walfish-Ikegami (W-I) path loss model is useful for urban areas and flat suburban areas which have regular building height. The Longley- Rice model gives accurate results over rough terrain since it does not account for foliage and buildings. The Hata-Okumura model has vast applications in a rural environment but its application is limited in urban (built up) areas.

TABLE III
COMPARATIVE ANALYSIS OF VARIOUS PROPAGATION MODELS

Model	Frequency Range	Type of Terrain	Pros	Cons
Okumura's Model	150MHz to 1950MHz	Urban, Suburban & Rural	Best Model for dense cities	Slow response to quick terrain profile changes and not good in rural areas
Hata Model	150MHz to 1500MHz	Urban, Sub-urban & Rural	Suitable for large cell mobile system	Not so good for rural hilly terrains
Hata-Okumura Model	150MHz to 1.5GHz	Metropolitan areas, small and medium sized cities, sub-urban and rural areas	Fairly accurate path loss estimation	Does not model propagation in cellular systems with higher frequencies
Stanford University Interim Model	2.5GHz to 2.7GHz	Hilly and Flat Terrains	Used to estimate the signal strength in all three terrain profiles like urban, suburban and rural.	Does not divide terrain profile into the rural, suburban and urban. This can be additional source of calculation incorrectness
Two ray Ground Reflection Model	Applicable to any Frequency	Plane earth	Useful and Simple model for propagation over long distances	Results are less satisfactory for shorter distances
Longley rice Model	20 MHz to 20GHz	Point to Point communication over different terrains	Path-specific parameters can be easily calculated when the detailed terrain profile is available	Does not provide a way of calculating correction due to environmental factor
COST 231 Hata Model	1500MHz to 2GHz	Urban, Sub-urban and Rural	Suited for predictions in microcells given that transmit antennas are above roof top	Graphical expressions are not practical
COST 231 Walfisch-Ikegami Model	800MHz to 2000MHz	Flat urban terrain	This model distinguishes between LoS and NLoS	More complex
Ericsson Model	1900MHz	Urban, Sub-urban and Rural	Very fast	Extensive test measurements are carried out in order to collect the data

IV. CONCLUSION

In this work, we surveyed different propagation path models to calculate path attenuation in different areas i.e. rural, sub-urban and urban. It is clearly observed that selection of an appropriate propagation model is critical in determining how many base stations are required to provide proper coverage for a network

ACKNOWLEDGMENT

We sincerely thank our chancellor and world-acclaimed spiritual leader, Dr. Mata Amritanandamayi Devi, popularly referred to as Amma, for the immense motivation, support and encouragement provided for this research work. This research is partially funded by Information Technology Research Agency (ITRA), Ministry of Electronics and Information Technology (MeitY), Government of India

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